

Ramakrishna Saravanan

M.S., Artificial Intelligence

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EDUCATION

Northwestern University

Master of Science in Artificial Intelligence (GPA: 3.975)

Evanston, IL

Sep 2025 – Dec 2026 (Expected)

Indian Institute of Technology Patna

B.Tech in Engineering Physics (CPI: 8.03)

Patna, India

July 2021 – May 2025

EXPERIENCE

Says

AI Engineer Intern, Team Auris

Evanston, USA

Apr 2026 – Present

- Architected the platform's foundational recommendation engine to scale across a catalog of **7.2M episodes** and **48K media collections**, driving the startup's transition towards targeted, user-centric feeds.
- Modelled multifaceted **user personas** via a multi-attribute vector search pipeline, dynamically blending metadata embeddings with explicit and implicit behavioral signals.
- Engineered a full-stack **React/FastAPI testing tool** to dynamically tune metadata weights for live recommendation simulation and quality auditing.

Samsung Research Institute Bangalore (PRISM)

Visual Intelligence Team, IIT Patna

Bangalore, India

July 2024 – Mar 2025

- Automated a manual content curation workflow for Samsung's movie catalog by engineering an on-device pipeline that extracts context aware aesthetic thumbnails in **less than 10%** of the video's total runtime.
- Engineered frame-pruning and chunking strategies to optimize multimodal inference under strict **CPU-only compute constraints**, cutting pipeline latency by **87.5%**.
- Integrated face detection, subtitle removal, and diverse keyframe selection algorithms, surpassing established image quality benchmarks like Hecate (**4.305 vs. 4.195**) and performing on par with human-selected targets (**3.273 vs. 3.326**).

SPARK Summer Research Internship, IIT Roorkee

Summer Research Intern, DDIR Lab

Roorkee, India

May 2024 – July 2024

- Engineered a semi-automated preprocessing and image segmentation framework using **OpenCV** to ingest **50,000** high-resolution CT scans, cutting manual annotation workflows by **90%** and saving 4–5 weeks of manual labor.
- Designed and optimized a **U-Net** image segmentation model using **Pytorch** for aerospace-grade manufacturing defect analysis, driving detection accuracy from a **78%** baseline to **97.3%**.
- Formulated a resolution-preserving fragmentation and data augmentation strategy to successfully isolate single-pixel sensor noise from structural voids, eliminating baseline failures in identifying critical large-scale defects.

PROJECTS

ProductVisionLM - Multimodal Instruction Tuning Pipeline

May 2026 – June 2026

- Engineered a multimodal vision-language model based on the LLaVA architecture, aligning a ViT-Large visual encoder with a Mistral 7B decoder via a fully trained GELU projection bridge.
- Executed parameter-efficient fine-tuning (LoRA) on a curated dataset of 50,000 image-text pairs from Amazon's Toys and Fashion subsets to autonomously generate vision-grounded product descriptions.
- Architected a multimodal LLM-as-a-Judge evaluation framework to rigorously verify the contextual alignment of generated outputs, establishing a 68% baseline accuracy in zero-shot visual instruction tasks.

AutoRL - Strategy Explorer.

Jan 2026 – Mar 2026

- Architected an agentic pipeline that converts unstructured rules into dynamic environments, utilizing a PPO model to extract optimized strategies and deliver actionable insights.
- Engineered an autonomous training architecture featuring an LLM feedback loop to self-evaluate inference, dynamically tune parameters, and execute continuous training without human intervention.

- Validated the framework’s scalability using Minesweeper as a spatial reasoning benchmark, autonomously deducing game rules to achieve a **70% win rate** over **10000** episodes.

Fallacy - Multimodal Claim Verification Engine

Jan 2026 – Mar 2026

- Engineered a browser extension to combat digital misinformation, delivering real-time claim extraction and evidence verification directly overlaid on YouTube videos.
- Architected a two-stage LLM pipeline utilizing Gemini and Search Grounding for zero-shot claim extraction, integrated with a VADER-based hybrid emotion-scoring system to evaluate rhetorical intensity.
- Evaluated the system’s zero-shot capabilities against a custom-curated dataset of **50** videos, achieving an **86%** accuracy rate in identifying, verifying, and scoring misleading claims.

Enterprise RAG System for Technical Documentation

Oct 2025 – Dec 2025

- Architected a scalable **RAG framework** using LangChain and OpenAI LLMs to generate accurate, context-grounded answers for enterprise documentation queries.
- Built a high-performance **dense retrieval pipeline** with semantic chunking, HuggingFace embeddings, and FAISS-based ANN search for fast and relevant document retrieval.

RESEARCH

Interpretability and Robustness in Machine Learning

July 2024 – May 2025

- Investigated interpretability of deep neural networks using saliency mapping on ResNet-based breast cancer classifiers, providing insights into the image regions driving model decisions.
- Evaluated model robustness under input perturbations and weight pruning, analyzing the trade-off between noise sensitivity, network sparsity, and classification performance.

Ethereum Phishing Scam Detection

Aug 2023 – Mar 2024

- Conducted experiments to analyze graph-based representations for fraudulent behavior detection and evaluate the effectiveness of GNNs over traditional classification approaches.
- Developed a Graph Convolutional Network (GCN) for Ethereum phishing account detection, leveraging transaction graph structures across 6.3 million nodes to achieve **98%** classification accuracy.

LEADERSHIP EXPERIENCE / CLUBS

- **Team Suspension Lead, IITP Motorsports** – Directed a student team in the end-to-end design and manufacturing of a Formula Student race car suspension system, focusing on optimizing overall vehicle dynamics
- **Mentor, Tamizh Sangam (IIT Patna)** – Mentored incoming students from Tamil Nadu through their academic and personal challenges, providing peer support and fostering a strong, inclusive community to ease their transition into university life
- **Student-Athlete, IITP Basketball & Tennis** – Represented the university in intracollegiate tournaments.

ACCOMPLISHMENTS

- **SPARK Fellowship '24**: Selected for the prestigious summer research program of IIT Roorkee in my junior year.
- **JEE MAINS 2021** – Achieved a **98th** percentile ranking, distinguishing myself among a national cohort of over a million candidates nationwide.
- **JEE Advanced 2021** – Secured an All India Rank (AIR) of **9,822**, placing among the highest percentile of elite engineering applicants in India.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C, C++.
- **Programming Frameworks:** PyTorch, TensorFlow, NumPy, Pandas, SciPy, Scikit-learn, Hugging Face Transformers/PEFT, LangChain, Ollama, vLLM, OpenCV, Scikit-Image, PIL.
- **Deployment Tools:** Git, Anaconda, MySQL, Docker, Azure
- **Web Development:** JavaScript, HTML, CSS